GE Healthcare

Installation Instructions

Helix Program
Technical Instruction
Document
ME_B

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Preface

This preface includes the following topics:

- Document information
- Formatting conventions
- Documentation set overview
- Intended use statement
- Safety information
- Service and support information
- User information
- Provisions

1.1 Document Information

This section covers the following topics:

- Revision history
- Referenced documentation

1.1.1 Revision History

The document's revision history is presented in a table on the document's copyright page, which is after the title page. It includes the revision number, date, author, and a description of the changes made to the document. In addition, the document and revision numbers are displayed at the foot of each page.

1.1.2 Referenced Documentation

The following documents are referenced within this service manual:

- Helix Technical Instruction: Configuration Manual (DOC0683569 Rev 2)
- Helix Technical Instruction: Content Manual (DOC0683573 Rev 2)

1.2 Formatting Conventions

The formatting conventions used in this document are described in this section. Topics include

- Typographic conventions
- Single-step procedures
- Type vs. enter
- Versioning variables

1.2.1 Typographic Conventions

Different typefaces are used to indicate different types of information, as described in Table 1.

Table 1: Typeface Conventions

Convention	Description
`Monospace type`	Code, commands, and text displayed on the terminal is presented in this font.
<*`variable`*>	Arrow brackets and italics indicate variable information. For example, <*`hostname`*> indicates that you need to enter the appropriate hostname.
Keystrokes	Keys on the keyboard are presented in bold. Letters are case sensitive. For example, type **y** and press **Enter**.
GUI elements	Menus and buttons in a GUI are presented in bold. For example, click **Finish**.

1.2.2 Single-Step Procedures

If a procedure has a single step, it is presented as follows:

- Enter the following command to create the file sample.txt:

```
vi sample.txt
```

1.2.3 Type vs. Enter

If you need to press **Enter** after typing a command, the convention is as follows:

- Enter the command `./init.sh`.

  This means to type `./init.sh` and press **Enter**.

If you do not need to press **Enter**, the convention is as follows:

- Type **y** at the prompt.

  This means to type the letter **y**; that's all. Remember that keystrokes are often case-sensitive.

1.2.4 Versioning Variables

The # sign is used to indicate variables in a product's version number. For example, use build number 1.0.0.0.###.

1.3 Documentation Set Overview

The Helix product documentation set is designed to help medical facility IT staff members to install, configure, understand, maintain, and troubleshoot the Helix system. Online help is available for technician use.

The full Helix product documentation set consists of the following documents:

- Release Notes
- Installation Manual
- Configuration Manual
- Content Manual
- Troubleshooting Manual
- Online Help (Operator's Guide)

1.3.1 Release Notes

Release notes contain the latest information pertaining to the release that is not contained in the other manuals. Topics include

- Product version
- How to contact customer support
- Document revision history
- Related documentation
- Prerequisite information
- Open issues
- Features working in the build
- Changes since the previous release
- Location of installation instructions
- Installation expectations
- Requested Issues
- Known issues

1.3.2 Installation Manual

The purpose of the Installation Manual is to provide all the information that is necessary to install the Helix product. Topics include

- Configuration
- Prerequisites
- Installation instructions
- Verification instructions

Post installation steps are contained in the Configuration and Maintenance Manual.

1.3.3 Configuration and Maintenance Manual

The purpose of the Configuration and Maintenance Manual is twofold:

- To provide the post-installation steps that are necessary to make the product fully functional
- To provide information that is necessary to perform routine maintenance

Topics include

- Post-installation configuration

  This topic describes how to make the installed product fully functional.
- Configuration reference

  This topic describes the configurable elements of the Helix product.

1.3.4 Troubleshooting Manual

The purpose of the Troubleshooting Manual is to provide information that is necessary to troubleshoot the Helix product. Topics include

- Error messages and their descriptions
- Descriptions of log files
- Procedures to resolve specific problems

1.3.5 Content Manual

The purpose of the Content Manual is to help users understand the Helix product. The Content Manual contains facts, concepts, and explanations to help users understand the architecture, functions, use, and components of the Helix product.

1.3.6 Online Help

Online help is available through the graphical user interface of the Helix client and serves as an Operator's Guide. Online help is accessed through the help button in the user interface.

Online help is also provided for users of the Terminology Authoring Tool.

1.4 Intended Use Statement

The components of the Helix healthcare information system are designed to support the clinical and administrative activities associated with patient care delivery in the various departments of either hospitals or integrated delivery networks for medical enterprises. Specifically, the Helix product helps to facilitate administrative and clinical workflow in the following areas:

- Patient tracking
- Clinical documentation
- Results entry
- Device interfacing
- Results reporting

The Helix product provides access to patients' electronic medical records, integrating information and data from departments and ancillaries across the enterprise.

The Helix product does not include any clinical diagnostic functions that require direct contact with a patient, such as taking measurements, performing calculations, or acquiring and storing images. Rather, it is to be used by various departments within a medical enterprise to facilitate the provisioning of care by providing access to electronic medical records.

Helix integrates with other Health Information Systems (HIS) and devices through standard HL7 interfaces and is intended to be used by technologists, clinicians, physicians, and other trained, site-authorized system users.

1.5 Safety Information

The Helix product stores data entered by clinical practitioners and makes computations based on the formulas provided by IT system administrators. The accuracy of data in the Helix product depends on the accuracy of data input.

The following topics are covered in this section:

- Potential electrical, mechanical, or physical hazards
- Tips, notes, and safety messages
- Auto-logoff and message boxes
- Security considerations

1.5.1 Potential Electrical, Mechanical, or Physical Hazards

Symbols that can appear on equipment are listed and defined in Table 2.

Table 2: Symbol Definitions

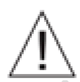
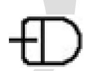
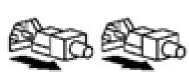
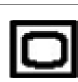
Symbol	Description
	Attention—Consult accompanying documents before using the equipment.
	SHOCK HAZARD—Indicates the presence of hazardous electrical circuits or electrical shock hazards.
	Keyboard port.
	Mouse port.
	USB port.
	Multiple power sources.
	VGA monitor port.
	Power on/standby switch.
	Internal component status LED. Red or amber indicates a malfunction; green indicates normal operation.

Table 2: Symbol Definitions

Symbol	Description
	Power supply status LED. Amber indicates a malfunction; green indicates normal operation.

1.5.2 Tips, Notes, and Safety Messages

The following list describes the various standard messages that can appear in the service manual:

TIP: Provides additional information about a procedure, such as a shortcut or an alternative method to complete a task.

NOTE: Provides additional information about a related topic. It does not pertain to hazards.

CAUTION: Indicates a potentially hazardous situation, which, if not avoided, may result in minor or moderate injury. The “Caution” message may also be used to alert against unsafe practices or to describe recommended practices for the safe and effective use of the Helix product.

WARNING: Indicates a potentially hazardous situation, which, if not avoided, could result in death or serious injury. This includes serious adverse reactions or limitations in use. Steps that must be taken to avoid or minimize the hazard will be included.

1.5.3 Auto-Logoff and Message Boxes

If a message is displayed when the system auto-logs off, the message will still be displayed when the next user logs on, regardless of whether it is the same or a different user. However, when the message box is dismissed by the user, a grey screen is displayed and the user must exit the system to continue.

1.5.4 Security Considerations

The Helix product provides two layers of security in accessing patient data:

- Authentication

  Authentication requires use of IDs and passwords.

- Authorization

  Authorization to access data is handled by role-based access control (RBAC). A user has privileges based on that user’s defined role and profile. A user’s access to patient data is controlled by defined work groups and their “legitimate relationship” to the patient.

NOTE: Failure to properly configure security access could result in unauthorized personnel gaining entry to the system, accessing privileged modules, and viewing or recording patient data.

1.6 Service and Support Information

The GE Healthcare Centricity Enterprise Support Center provides day-to-day support for product issues. The Support Center staff (Tier 3) acts as secondary support to the trained IT staff at our customer sites (Tier 2).

Table 3: GE Healthcare Support Center

Support Center Hours	Contact Information
Business Hours Support: 6 A.M. to 5 P.M., Pacific Time, Monday through Friday, excluding GE Healthcare holidays.	877-946-2277 Submit non-critical issues through the eService VanWeb portal.
Off-Hours Support for Critical Issues: Critical production issue support is provided 24 hours a day, 7 days a week, including holidays.	206-621-1044, Option 1

1.6.1 Service and Support Web Addresses

You can also report problems directly into Vantive via VanWeb on the Internet. (https://eservice.gehealthcare.com/)

1.7 Provisions

Provisions included in this section are

- Customer responsibility for medical records
- Data collection
- Design changes
- Patient care
- Unintended references
- Document reproduction
- Disclaimer

1.7.1 Customer Responsibility for Medical Records

Customers are responsible for determining the form, type, and presentation of the data and documentation that constitute the patient medical record. The Helix product enables customers to configure the type of data captured and the electronic and printed displays of data. It is the customer's responsibility to become familiar with the features of the system to make informed decisions about the data, processes, media, and formats that the customer may choose to capture, extract, archive, or display for the purposes of an official medical record.

1.7.2 Data Collection

The Helix product, though it may interface with patient-monitoring equipment, is not intended to function as a replacement for the way patient information is obtained from that equipment; rather, the Helix product must be employed as a supplemental device that allows the clinical staff to enter, store, retrieve, and view data in an efficient and flexible way.

1.7.3 Design Changes

Due to design changes associated with continuing product improvements, information in this manual is subject to change. GE Healthcare reserves the right to change software design at any time, including changes that can impact this manual. GE Healthcare assumes no responsibility for any errors or inconsistencies appearing in this manual that result from product design changes or upgrade.

NOTE: Users must assume that any printed form of the Helix product documentation set must necessarily be out-of-date unless verified against the latest electronic version available from GE Healthcare.

1.7.4 Patient Care

Operation of the Helix product should neither circumvent nor take precedence over required patient care, nor should it impede the human intervention of attending nurses, physicians, or other medical personnel in a manner that would have a negative impact on patient health.

1.7.5 Reference to Persons, Places, and Institutions

References to persons, places, and institutions used within this manual are solely intended to help the reader understand how to use the Helix product. Extreme care has been taken to use fictitious names and related information in the examples and illustrations provided herein. Any similarities to persons either living or dead or with current or previously existing medical institutions must be regarded as coincidental.

1.7.6 Document Reproduction

No part of this manual may be reproduced, stored in a retrieval system, or transmitted in any form or by any means—electronic, mechanical, photographic, etc.—without the prior written permission of the General Electric Company. All inquiries regarding this manual or the Enterprise Clinical Information System must be directed to GE Healthcare HCIT.

1.7.7 Disclaimer

GE Healthcare carries no liability for any damages to property or life caused by wrong, and/or untested alert settings made by the customer.

Chapter 2: Installing Helix on Two Servers in a Multi-Tier Configuration

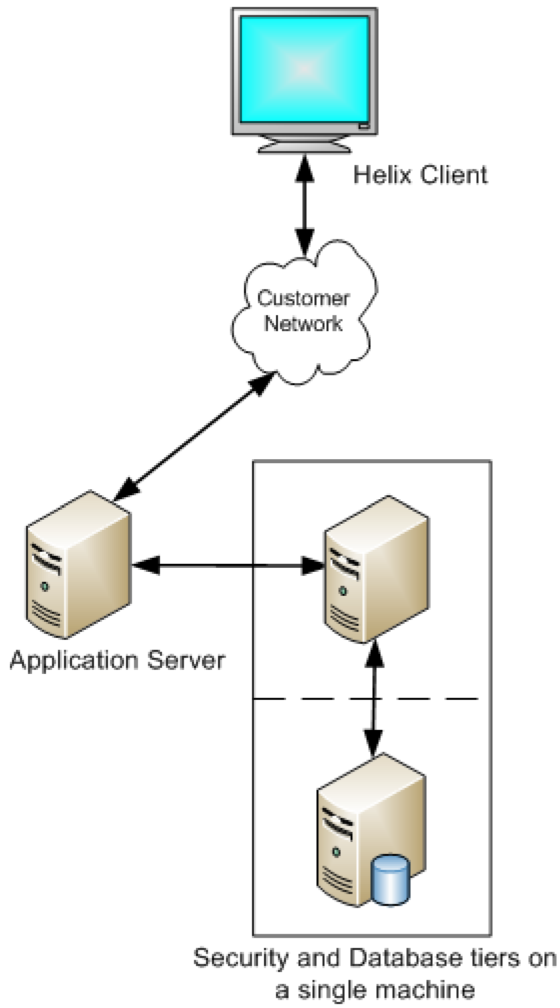
This chapter includes information on the following topics:

- Overview
- Prerequisites
- Installing to the database server
- Installing to the security server
- Installing to the application server
- Post installation
- Manually configuring the network

GE Healthcare

2.1 Overview

This chapter shows you how to install the Helix product in a multi-tier configuration using two physical servers. This Helix multi-tier configuration is shown in the figure below:



You will be installing to the database/security server, the application server, and the client. At a high level, installation follows the steps below:

1. Install to the database server:
   a. Install the operating system.
   b. Configure the network.
   c. Install the database software.
   d. Verify that the installation was successful.
2. Install to the security server:
   a. Install the security software.

   b. Verify that the installation was successful.
3. Install to the application server:
   a. Install the operating system.
   b. Configure the network.
   c. Install the application software.
   d. Verify that the installation was successful.

NOTE: This release is not for patient use.

2.2 Prerequisites

This section lists what is required to install the Helix product in a multi-tier configuration using two physical servers.

2.2.1 Hardware Requirements

For the database/security server:

- The vendor and model of the server must be in Red Hat's HCL for RHEL version 5.4: the vendor support matrix or HCL also must show support for RHEL 5.4.
- CPU: Intel L5520 or better.
- Memory: 12GB or greater.
- Disk: 3 x 73GB SAS 15K 2.5" drives with hardware controller to support 2 drives RAID1 mirror and the third as a hot-spare.
- Other: 2 x QLogic 4Gb FC single-port PCIe HBA.
- Other: 2 x integrated 1GB/sec integrated NICs.
- Other: N + 1 Power supplies (normally a total of two power supplies for redundancy).

For the application server:

- The vendor and model of the server must be in Red Hat's HCL for RHEL version 5.4: the vendor support matrix or HCL also must show support for RHEL 5.4.
- CPU: Intel L5520 or better.
- Memory: 6GB or greater.
- Disk: 2 x 73GB SAS 15K 2.5" drives (RAID1 mirror).
- Other: 2 x QLogic 4Gb FC single-port PCIe HBA.
- Other: 2 x integrated 1GB/sec integrated NICs.
- Other: N + 1 Power supplies (normally a total of two power supplies for redundancy).

2.2.2 Network and System Settings

You'll need the following information to configure the network:

- **Hostname** for both servers
- **IP address** for both servers*
- **Netmask** for both servers*
- **Gateway** for both servers*
- **DNS server** for both servers*
- **jdbc hostname** for the database server

- **jdbc port** for the database server
- **Security hostname** for the security server
- **Monitoring hostname** (Disabled. Use "localhost.")

* This value isn't required if using DHCP to set up the network.

2.3 Installing to the Database Server

This section includes the following information:

- Overview
- Installing to the database server
- Validating the installation on the database server

2.3.1 Overview

The installation can be broken down into three steps:

1. Install the operating system.
2. Configure the network.

   NOTE: You can set up the network two ways: using the deployment script or setting it up manually. We recommend you use the deployment script.

3. Run the deployment script.

2.3.2 Installing to the Database Server

1. Insert the Database Installation disk into the drive of the host server.

   The boot menu appears.

2. Type `helix` and press **ENTER**.
3. When the installation is finished, the system will reboot.
4. (Optional) Configure the network manually.
   a. Follow the procedure in "(Optional) Manually Configuring the Network" on page 25 (2.7), then proceed to step b.
   b. Skip step 5 and proceed to step 6.
5. Enter the following command to configure the network:

   ```
   hcit-deploy -n
   ```

   You will be prompted for the following values:

   - **Hostname** of the application server
   - **IP address** of the application server
   - **Netmask** of the application server
   - **Gateway** of the application server
   - **DNS server** of the application server

6. Enter the following command to install the Helix and database components:

   ```
   hcit-deploy -s
   ```

7. Insert the installation DVD when prompted, if necessary.
8. Press **ENTER**.
9. Type **n** when prompted to install the monitoring client.

   When the installation is finished, the message “Complete!” appears and you are returned to the system prompt.

2.3.3 Validating the Installation on the Database Server

- Enter the following command to verify that the database is running:

  ```
  service oracle status
  ```

  If the database installed correctly, you’ll see a message that the database is running.

2.4 Installing the Security Software

This section includes the following information:

- Overview
- Installing the security software
- Validating the installation of the security software

2.4.1 Overview

Since the operating system has already been installed on this server and the database service exists, you are ready to run the deployment script.

2.4.2 Installing the Security Software

1. Insert the Security Installation disk into the **same drive as the database installation.**
2. Open the following file for editing:
   `/etc/ecis/ecis-release`
3. Find the following text in the file:
   `tiername=helix_dbms_x86_64`
4. Change "dbms" to "security."
5. Save and close the file.
6. Enter `hcit-deploy -s`.
7. When prompted, enter the following values:
   - For the jdbc hostname, enter **localhost**.
   - For the jdbc port, enter **1521**.
   - For the monitoring host, enter **localhost**.

   NOTE: The monitoring host is bypassed.
8. Insert the Security Installation disk when prompted, if necessary.
9. Press **ENTER**.
10. Type **n** when prompted to install the monitoring client.

    When the installation is finished, you will be returned to the system prompt.

2.4.3 Validating the Installation of the Security Software

To validate the installation:

1. Open Internet Explorer.
2. Navigate to https://*<IP address of the security server>*:7010/asi.
3. Log on using "admin" as the username and "password" as the password.
4. In the left column, expand **Identity > User.**
5. In the middle panel, verify that "testuser" and "weblogic" are listed.

2.5 Installing to the Application Server

This section includes the following information:

- Overview
- Installing the application software
- Validating the installation on the application server

2.5.1 Overview

The installation can be broken down into three steps:

1. Install the operating system.
2. Configure the network.
3. Run the deployment script.

2.5.2 Installing the Application Software

1. Insert the Application Installation disk into the drive of the application server.

   The boot menu appears.
2. Type `helix` and press **ENTER**.

   The system will automatically reboot when finished.
3. (Optional) Configure the network manually.
   a. Follow the procedure in "(Optional) Manually Configuring the Network" on page 25 (2.7), then proceed to step b.
   b. Verify that the hostname of the server is the same as contained in the `/etc/sysconfig/network` file.
   c. Verify that the application server hostname is resolved to the IP address.
   d. Verify that the format of the `/etc/hosts` file is as follows:

      `<IP address> <full hostname> <short hostname>`
   e. Skip step 4 and go to step 5.
4. Enter `hcit-deploy -n` to set up the network.

   You will be prompted for the following values:
   - **Hostname** of the application server
   - **IP address** of the application server
   - **Netmask** of the application server
   - **Gateway** of the application server
   - **DNS server** of the application server

5. Enter `hcit-deploy -s`.

   You will be prompted for the following values:

   - **jdbc hostname** for the database server
   - **jdbc port** for the database server
   - **Security hostname** for the security server
   - **Monitoring hostname** (Disabled. Use “localhost.”)

6. Insert the Application Installation DVD when prompted, if necessary.
7. Press **ENTER**.
8. Type **n** when prompted to install the monitoring client.
9. Type **y** when prompted to deploy the EAR.

   A message that the system is ready will appear after the installation is finished.

2.5.3 Validating the Installation on the Application Server

To validate the installation:

1. Open Internet Explorer.
2. Navigate to https://*<IP address of the application server>*:9001/console.
3. Log on using “weblogic” as the username and password.
4. In the left column, click **Deployment**.
5. Verify the following:
   - Status: **Active**
   - Health: **OK**

2.6 Post Installation

The next steps are to create an Admin user and install the client.

- To create an Admin user, see Chapter 3: Creating an Admin User for Centricity Framework.
- To install the client, see Chapter 4: Installing the Helix Client.

2.7 (Optional) Manually Configuring the Network

You can configure the network using a static IP address or DHCP.

2.7.1 Using a Static IP Address to Configure the Network

1. Log on to the system as root.
2. Open the `/etc/sysconfig/network` file for editing.
3. Edit the file to include the hostname and domain of the database server.

   For example,

   ```
   NETWORKING=yes
   NETWORKING_IPV6=no
   HOSTNAME=myhostname.mydomain.com
   ```
4. Save and close the file.
5. Open the `/etc/sysconfig/network-scripts/ifcfg-eth0` file for editing.
6. Change `ONBOOT=no` to `ONBOOT=yes`.
7. Edit the file using the network settings of the database server:

   ```
   IPADDR=<your IPaddress>
   NETMASK=<your netmask>
   GATEWAY=<your gateway address>
   DNS1=<your DNS server address>
   DNS2=
   DNS3=
   ```
8. Save and close the file.
9. Open the `/etc/hosts` file for editing.
10. Add the IP address, hostname.domain, and alias hostname of the database server to the end of the file.

    For example,

    ```
    3.28.56.202   myhostname.mydomain.com   myhostname
    ```
11. Save and close the file.

    The network configuration is now complete.
12. Activate the network with the following command:

    ```
    /sbin/service network restart
    ```
13. Type `exit` and press **ENTER** to log off.
14. Log back on to the system as root.
15. Verify network functionality by pinging your gateway address with the following command:

```
ping <your_gateway_address>
```

This command will successfully return pings if the configuration is setup correctly and the machine is correctly connected to the network.

TIP: Press **Ctrl+C** to stop the ping command.

2.7.2 Using DHCP to Configure the Network

1. Log on to the system as root.
2. Open the `/etc/sysconfig/network` file for editing.
3. Edit the file to include the hostname and domain of the database server.

   For example,

   ```
   NETWORKING=yes
   NETWORKING_IPV6=no
   HOSTNAME=myhostname.mydomain.com
   ```
4. Save and close the file.
5. Open the `/etc/sysconfig/network-scripts/ifcfg-eth0` file for editing.
6. Change `ONBOOT=no` to `ONBOOT=yes`.
7. Edit the file as follows:

   ```
   BOOTPROTO=dhcp
   TYPE=Ethernet
   ```
8. Save and close the file.

   The network configuration is now complete.
9. Activate the network with the following command:

   ```
   /sbin/service network restart
   ```
10. Type `exit` and press **ENTER** to log off.
11. Log back on to the system as root.
12. Verify network functionality by pinging your gateway address with the following command:

    ```
    ping <your_gateway_address>
    ```

    This command will successfully return pings if the configuration is setup correctly and the machine is correctly connected to the network.

    TIP: Press **Ctrl+C** to stop the ping command.

2.7.3 (Optional) Configuring NTP Services to Configure the Setup Clock

If you want to configure the setup clock, do the following:

1. Open the `/etc/ntp.conf` file for editing.
2. Edit each line that starts with `#server` to include the appropriate NTP server information of the database server.

   NOTE: Each line should refer to only one server. If there is more than one server, add additional lines for each server's settings. Ensure that each line ends with "iburst."
3. Remove the # sign from each of the edited server lines.
4. Save and close the file.
5. Enter `service ntpd stop` to stop the service.
6. Enter `ntpdate <ntp server>`, substituting the host name or IP address of the NTP server.
7. Enter `service ntpd start` to start the service.

Chapter 3: Creating an Admin User for Centricity Framework

This chapter covers the following topics:

- Creating an Admin User
- Assigning a user to a group

3.1 Creating an Admin User

1. Open an Internet browser.
2. Navigate to https://<*your Server DNS*>:7010/asi/.

   Make sure you include the "s" in "https."
3. Enter the username **Admin** and the password **Password**.

   NOTE: If a security dialog prompts you to continue, click **Yes**.
4. Select **Administration Console > Identity > Users** in the left panel.
5. Click **New**.
6. Enter **Admin** in the name field and click **OK**.
7. Click **Edit**.
8. Click **Set Password**.
9. Enter **Password** in the password field and confirm it.
10. Click **OK**, then click **OK** again.

3.2 Assigning a User to a Group

1. After adding the user "Admin," click **Assign**.
2. Select **Administrators** in the Available Groups list and click **<<Add**.
3. Click **Close**.
4. Log off OES Admin Console.

The next step is to install the client. For information, see Chapter 4: Installing the Helix Client.

Chapter 4: Installing the Helix Client

This chapter includes the following topics:

- Prerequisites
- Installing to the Helix client
- Configuring Centricity Framework

4.1 Prerequisites

For the client:

- Machine type: laptop/desktop
- CPU: Pentium class or equivalent
- Processor speed: 1 Gigahertz
- Memory: 1 gb
- Video capabilities: minimum resolution of 1024 x 768 Pixels
- Monitor size: 19-inch
- Network: 100 mb
- Operating system: Windows XP
- Runtime environment: .NET Framework 3.5 SP1

4.2 Installing to the Client

Installing Helix takes two steps:

1. Install the Helix client.
2. Configure the network.

4.2.1 Installing the Helix the Client

To install the Helix client:

1. Insert the Client Installation disk.
2. Navigate to [root]\client\clientfiles\setup_1.0.0.0.193.exe\.
3. Double-click the setup.exe file.

   The InstallShield wizard appears.
4. Click **Next**.
5. Accept the terms of the license agreement, then click **Next**.
6. Enter a username and organization name.
7. Click **Next**.
8. Select where you want to place the application shortcuts.

   The options are Desktop, Program Menu, and Startup.
9. Click **Install**.

   Status bars show the progress of the installation.
10. Click **Finish**.

4.2.2 Configuring the Network

To configure the network:

1. Open the following file for editing:

   ```
   C:\Program Files\GE Healthcare\Centricity
   Enterprise\Client Workstation\1.0\bin\CF.exe.config
   ```
2. Replace the server name (shown in bold) with the name of your application server.

   ```
   <setting name="ServerName" serializeAs="String">
   <value>localhost</value>
   </setting>
   ```
3. Save and close the file.
4. Open the following file for editing:

```
C:\Program Files\GE Healthcare\Centricity
Enterprise\Client Workstation\1.0\bin\
Client.Foundation.dll.config
```

5. Replace the value of CEServer.URL.Address with the name of your application server:

```
<appSettings>

<add key="CEServer.URL.Address" value="<application
server>"/>

<add key="CEServer.URL.Port" value="8010"/>
```

6. Save and close the file.

   You can start the application from any of the shortcuts you selected during install or double-click CF.exe in the installation directory. See your system adminstrator for logon requirements.

4.3 Configuring Centricity Framework

1. Launch the Centricity Framework client.
2. Login using the credentials Admin | Password.
3. Click **Import/Export** in the vertical toolbar.
4. Select **Import** for the Action.
5. Select **Read from file** for the Source.
6. Enter install_helix.xml for the Source File Name
7. Click **Import** in the top right corner
8. Click **Log Off**.

Appendix

This appendix covers the following topics:

- URL reference
- Creating a bonded network interface

A.1 URL Reference

Table 1 lists the application URLs.

Table 1: URL Reference

Application	Listening on Port . . .	URL
Add users/roles/policies	7010	https://localhost:7010/asi
ESBServer1 Server JMX Port	8012	
SPServer1 Server JMX Port	9012	
ESBServer1 Admin	http://<*your server IP*>:8001/ console	
ESBServer1 Admin SSL PORT	https://<*your server IP*>:8002	
ESBServer1 Server Port	http://<*your server IP*>:8010	
ESBServer1 Server SSL port	https://<*your server IP*>:8011	
SPServer1 Admin	http://<*your server IP*>:9001	
SPServer1 Admin SSL PORT	https://<*your server IP*>:9002	
SPServer1 Server Port	http://<*your server IP*>:9010	
SPServer1 Server SSL port	https://<*your server IP*>:9011	

A.2 Creating a Bonded Network Interface

After the operating system has been installed and before you install any Helix software, you can add availability by creating a bonded network interface. The instructions that follow show you how to create a bonded interface on an Ethernet interface and assign an IP address to it. Additional information can be found in Red Hat's Deployment Guide for RHEL 5.

Creating a bonded network interface requires that all commands be run with root permissions and that the interfaces to be bonded are physically connected to the network.

You will need the following information (the values shown are examples only):

- Bonded interface ID: bond0
- IP Address: 10.10.10.5
- Subnetmask: 255.255.255.0
- Gateway: 10.10.10.1
- All commands must be run with root permissions.
- Interfaces to be bonded must be physically connected to the network.

To create a bonded network interface:

1. Enter the following command to verify that the driver for the network interfaces supports the MII tool:

   ```
   ethtool eth0|grep Link
   ```

   The MII tool is used to verify that the network interface card is active. If successful, you will see "Link detected: yes."
2. Open the /etc/modprobe.conf file for editing.
3. Add the following line to the file:

   ```
   alias bond0 bonding
   ```
4. Save and close the file.
5. Create the file /etc/sysconfig/network-scripts/ifcfg-bond0 and open it for editing.
6. Add the following text to the file:

   ```
   ONBOOT=yes
   IPADDR=10.10.10.5
   NETMASK=255.255.255.0
   GATEWAY=10.10.10.1
   USERCTL=no
   ```
7. Save and close the file.
8. Open the /etc/sysconfig/network-scripts/ifcfg-eth0 file for editing.

9. Change the file to match the following:

   ```
   DEVICE=eth0
   BOOTPROTO=none
   ONBOOT=yes
   MASTER=bond0
   SLAVE=yes
   USERCTL=no
   ```

10. Save and close the file.
11. Open the /etc/sysconfig/network-scripts/ifcfg-eth1 file for editing.
12. Change the file to match the following:

   ```
   DEVICE=eth1
   BOOTPROTO=none
   ONBOOT=yes
   MASTER=bond0
   SLAVE=yes
   USERCTL=no
   ```

13. Save and close the file.
14. Reboot the system.

    You've successfully created a network bonded interface.
15. (Optional) You can verify that the bonding is working by doing the following:
    a. From another system, use the ping command to verify that the bonded server is up.
    b. Take down one of the bonded network interfaces by entering either of the following commands:
       - `/sbin/ifconfig eth0 down`
       - `/sbin/ifconfig eth1 down`
    c. Enter `/bin/cat/proc/net/bonding/bond0`.